

FACULDADE DE CIÊNCIAS DA UNIVERSIDADE DO PORTO
FACULDADE DE ENGENHARIA DA UNIVERSIDADE DO PORTO

Inference-Time Techniques for Open-Weight Language Models in Negotiation Games

Adriano Alexandre dos Santos Machado

WORKING VERSION



Master in Artificial Intelligence

Supervisor: Prof. Gil Rocha

Second Supervisor: Prof. Henrique Lopes Cardoso

June 5, 2026

Resumo

Este documento ilustra o formato a usar em dissertações na Faculdade de Engenharia da Universidade do Porto (FEUP). São dados exemplos de margens, cabeçalhos, títulos, paginação, estilos de índices, etc. São ainda dados exemplos de formatação de citações, figuras e tabelas, equações, referências cruzadas, lista de referências e índices. Este documento não pretende exemplificar conteúdos a usar. É usado o *Loren Ipsum* para preencher a dissertação. Lorem ipsum dolor sit amet, FEUP consectetur adipiscing elit. Etiam vitae quam sed mauris auctor porttitor. Mauris porta sem vitae arcu sagittis facilisis. Proin sodales risus sit amet arcu. Quisque eu pede eu elit pulvinar porttitor. Maecenas dignissim tincidunt dui. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Donec non augue sit amet nulla gravida rutrum. Vestibulum ante ipsum primis in faucibus orci luctus et ultrices posuere cubilia Curae; Nunc at nunc. Etiam egestas. Donec malesuada pede eget nunc. Fusce porttitor felis eget mi mattis vestibulum FEUP. Pellentesque faucibus. Cras adipiscing dolor quis mi. Quisque sagittis, justo sed dapibus pharetra, lectus velit tincidunt eros, ac fermentum nulla velit vel sapien. Vestibulum sem mauris, hendrerit non, feugiat ac, varius ornare, lectus. Praesent urna tellus, euismod in, hendrerit sit amet, pretium vitae, nisi. Proin nisl sem, ultrices eget, faucibus a, feugiat non, purus. Etiam mi tortor, convallis quis, pharetra ut, consectetur eu, orci. Vivamus aliquet. Aenean mollis fringilla erat. Vivamus mollis, purus at pellentesque faucibus, sapien lorem eleifend quam, mollis luctus mi purus in dui. Maecenas volutpat mauris eu lectus. Morbi vel risus et dolor bibendum malesuada. Donec feugiat tristique erat. Nam porta auctor mi. Nulla purus. Nam aliquam.

Palavras-chave: keyword1 • keyword2 • keyword3 • keyword4

Abstract

Here goes the abstract written in English. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Sed vehicula lorem commodo dui Faculdade de Engenharia da Universidade do Porto (FEUP). Fusce mollis feugiat elit. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Donec eu quam. Aenean consectetur odio quis nisi. Fusce molestie metus sed neque. Praesent nulla. Donec quis urna. Pellentesque hendrerit vulputate nunc. Donec id eros et leo ullamcorper placerat. Curabitur aliquam tellus et diam. Ut tortor. Morbi eget elit. Maecenas nec risus. Sed ultricies. Sed scelerisque libero faucibus sem FEUP. Nullam molestie leo quis tellus. Donec ipsum. Nulla lobortis purus pharetra turpis. Nulla laoreet, arcu nec hendrerit vulputate, tortor elit eleifend turpis, et aliquam leo metus in dolor. Praesent sed nulla. Mauris ac augue. Cras ac orci. Etiam sed urna eget nulla sodales venenatis. Donec faucibus ante eget dui. Nam magna. Suspendisse sollicitudin est et mi. Fusce sed ipsum vel velit imperdiet dictum. Sed nisi purus, dapibus ut, iaculis ac, placerat id, purus. Integer aliquet elementum libero. Phasellus facilisis leo eget elit. Nullam nisi magna, ornare at, aliquet et, porta id, odio. Sed volutpat tellus consectetur ligula. Phasellus turpis augue, malesuada et, placerat fringilla, ornare nec, eros. Class aptent taciti sociosqu ad litora torquent per conubia nostra, per inceptos himenaeos. Vivamus ornare quam nec sem mattis vulputate. Nullam porta, diam nec porta mollis, orci leo condimentum sapien, quis venenatis mi dolor a metus. Nullam mollis. Aenean metus massa, pellentesque sit amet, sagittis eget, tincidunt in, arcu. Vestibulum porta laoreet tortor. Nullam mollis elit nec justo. In nulla ligula, pellentesque sit amet, consequat sed, faucibus id, velit. Fusce purus. Quisque sagittis urna at quam. Ut eu lacus. Maecenas tortor nibh, ultricies nec, vestibulum varius, egestas id, sapien. Donec hendrerit. Vivamus suscipit egestas nibh. In ornare leo ut massa. Donec nisi nisl, dignissim quis, faucibus a, bibendum ac, diam. Nam adipiscing hendrerit mi. Morbi ac nulla. Nullam id est ac nisi consectetur commodo. Pellentesque aliquam massa sit amet tellus. Vivamus sodales aliquam leo.

Keywords: keyword1 • keyword2 • keyword3 • keyword4

Acknowledgements

Aliquam id dui. Nulla facilisi. Nullam ligula nunc, viverra a, iaculis at, faucibus quis, sapien. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Curabitur magna ligula, ornare luctus, aliquam non, aliquet at, tortor. Donec iaculis nulla sed eros. Sed felis. Nam lobortis libero. Pellentesque odio. Suspendisse potenti. Morbi imperdiet rhoncus magna. Morbi vestibulum interdum turpis. Pellentesque varius. Morbi nulla urna, euismod in, molestie ac, placerat in, orci.

Ut convallis. Suspendisse luctus pharetra sem. Sed sit amet mi in diam luctus suscipit. Nulla facilisi. Integer commodo, turpis et semper auctor, nisl ligula vestibulum erat, sed tempor lacus nibh at turpis. Quisque vestibulum pulvinar justo. Class aptent taciti sociosqu ad litora torquent per conubia nostra, per inceptos himenaeos. Nam sed tellus vel tortor hendrerit pulvinar.

Fusce gravida placerat sem. Aenean ipsum diam, pharetra vitae, ornare et, semper sit amet, nibh. Nam id tellus. Etiam ultrices.

Adriano Machado

Declaration of honour

I declare that this dissertation is my own work and has not previously been submitted or assessed at this or any other institution. References to other authors (statements, ideas, thoughts), as well as the use of published works, strictly comply with the rules of attribution and are duly identified in the text and in the bibliographic references, in accordance with the applicable referencing standards. I am aware that the practice of plagiarism or self-plagiarism constitutes an academic offence, as established in the U.Porto Code of Ethics.

I also declare that I have used Artificial Intelligence tools to complete parts of this dissertation, and that this use and its results are duly noted and have been carefully checked for errors, inaccuracies, unfounded attributions or other forms of bias in content and arguments, as well as determining authorship.

Adriano Alexandre dos Santos Machado



“We make a living by what we get, but we make a life by what we give.”

Winston Churchill

Contents

1	Introduction	1
2	Literature Review	3
2.1	Language Models	3
2.2	Inference-Time Reasoning Techniques	3
2.2.1	Chain-of-Thought Reasoning	4
2.2.2	Self-Correction	5
2.3	Multi-Agent Systems	5
2.4	Interaction Scenarios	6
2.4.1	Cooperative	6
2.4.2	Adversarial	7
2.4.3	Mixed	8
2.4.4	Team-Based Negotiation	8
3	Methodology	9
3.1	NegotiationArena Framework	9
3.1.1	Games	9
3.1.2	Evaluation Metrics	9
3.2	Models and Infrastructure	9
3.3	Benchmarking Open-Weight LLMs	9
3.4	Self-Refine	9
3.5	Team Negotiation	9
4	Results and Discussion	12
4.1	Benchmarking Open-Weight LLMs	12
4.2	Self-Refine in Multi-Agent Negotiation	12
4.3	Team Negotiation	12
	References	13
A	Inference-Time Technique Prompts	17
A.1	Self-Refine Prompts	17
A.1.1	Feedback Prompt	17
A.1.2	Refine Prompt	18
A.2	Negotiation Team Prompts	18
A.2.1	Discussion Prompt (Phase 2)	18
A.2.2	Ranking Prompt (Phase 3)	19
A.2.3	Reason-Laundering Prompt (Phase 3)	20

B	NegotiationArena Game Prompts	21
B.1	Buy-Sell Game	21
B.2	Trading Game	23
B.3	Ultimatum Game	25
B.4	Persona Prompts	27

List of Figures

2.1	Example of the Self-Refine process. The model critiques its initial output to generate an improved final response based on self-generated feedback.	5
2.2	Evolution of MAS paradigms. Adapted from Nguyen-Thi-Mai et al. [25].	6
3.1	Self-Refine process. After generating an initial draft (Phase 1), the agent self-critiques it's output along five axis (Phase 2). Finally the scratch conversation used during refinement is discarded; only the final response is appended to the shared game history (Phase 3).	10
3.2	Team deliberation process. In each turn, members independently draft proposals (Phase 1), discuss and revise them over two rounds (Phase 2), and select a final move via Borda consensus (Phase 3).	11

List of Tables

2.1	Comparison of different models. For Mixture of Experts (MoE) models, parameters are denoted as <i>Total / Active</i>	4
3.1	Open-weight models used across all experiments. Gemma, Qwen, and Mistral-Small are loaded with 8-bit quantization (bitsandbytes); the Ministral models are released already quantized to FP8 and are used at that native precision. VLM models are loaded with a vision-language (image-text-to-text) head; Qwen models run with thinking mode disabled.	9
B.1	Persona prompts appended to Player 2's system message in the persona experiments. An empty cell indicates the <i>default</i> condition (no text added).	27

List of Acronyms

FEUP Faculdade de Engenharia da Universidade do Porto

Chapter 1

Introduction

Almost everything in life depends on negotiation. From the moment we are born, we negotiate for attention and care. Developed in part through the unconscious imitation of those around us [12], negotiation requires the ability to infer others’ intentions and to act under uncertainty.

Large language models (LLMs) have demonstrated impressive capabilities across a number of tasks, from code generation to creative writing and question answering. As these models are increasingly deployed in agentic scenarios—where they must interact with other agents or humans to accomplish goals—negotiation appears as a particularly interesting benchmark.

Bianchi et al. [2] introduced NegotiationArena, an open-source framework for evaluating the negotiation abilities of LLM agents across scenarios such as the Ultimatum Game, resource trading, and price negotiation. Although highly valuable, the original evaluation was limited to closed-source models (GPT-4, GPT-3.5, Claude 2.1, and Claude 2), some of which are no longer available. This limitation raises concerns regarding reproducibility and accessibility.

Techniques such as Chain-of-Thought (CoT) allow models to think step-by-step, dividing complex problems into manageable sub-problems to reduce errors and improve interpretability. Similarly, Self-Correction mechanisms (or Self-Refine) enable models to critique and improve their own outputs through an iterative process of drafting, feedback, and refinement. While these emergent abilities have demonstrated success in domains like mathematics and coding, their specific impact on negotiation has not yet been fully studied.

Beyond individual reasoning, multi-agent collaboration offers another area for improvement. Cooperative debate, where multiple agents generate, critique, and refine each other’s responses, has been shown to improve factual accuracy and reasoning quality. In real-world negotiation, decisions are rarely made by a single individual; corporate acquisitions, policy agreements, and legal settlements typically involve teams that deliberate internally before engaging with the opposing side. This observation motivates the question of whether team-based negotiation — where each side is represented by a group of LLM agents that debate before responding — can lead to stronger negotiation outcomes.

This thesis investigates whether inference-time reasoning techniques and multi-agent collaboration can improve the negotiation capabilities of large language models. The work is organized

around three main contributions:

- How do open-source models compare to proprietary models in competitive negotiation scenarios?
- As inference-time reasoning techniques have shown promise individually, to what extent do they affect outcomes in multi-agent negotiation contexts?
- Can team-based negotiation, where each side is represented by a group of LLM agents that deliberate before responding, lead to stronger negotiation outcomes?

The remainder of this document is organized as follows. Chapter 2 provides a literature review covering the evolution of language models, inference-time reasoning techniques, multi-agent systems, and the different interaction scenarios relevant to this work. Chapter 3 presents the detailed work plan for the upcoming months.

Chapter 2

Literature Review

2.1 Language Models

The origins of language modeling date back to Shannon [31], which modeled natural language as a stochastic process. Building on this, Bengio et al. [1] proposed one of the first Neural Language Models (NLMs) estimating n-gram probabilities using neural networks. Thereafter, NLMs based on Long Short-Term Memory (LSTM) became the standard for many natural language tasks, including machine translation, text generation, and classification [24, 34].

The introduction of the Transformer architecture [43] addressed some limitations of LSTMs, allowing parallelization during training and capturing long-range dependencies through the self-attention mechanism. Applying this architecture, Radford and Narasimhan [29] pre-trained a language model on the next-word prediction task and showed that, when followed by task-specific fine-tuning, it achieved state-of-the-art performance on common-sense reasoning, question answering and many other tasks. GPT-2 [30] demonstrated that sufficiently large language models can perform downstream tasks such as summarization, translation, and question answering without task-specific fine-tuning.

The increase in the number of parameters, dataset size, and training compute of language models continued, supported by the power-law relationship detected in Kaplan et al. [18], which showed that an increase in these three components led to an improvement in performance. As models grew, so did the computational costs and risks associated with their deployment, which led the majority of frontier labs to keep the model weights proprietary; nevertheless, some have opted for an open-weight release (e.g., LLaMA [13], Mistral [16], DeepSeek [9]). Table 2.1 compares some of the main families of models.

2.2 Inference-Time Reasoning Techniques

Beyond simple improvements in performance, scaling also brought new capabilities that could not be predicted by extrapolating the performance of smaller models. Wei et al. [45] refers to them as *emergent abilities*. Among them are the ability to learn from examples (Few-Shot prompting,

Table 2.1: Comparison of different models. For Mixture of Experts (MoE) models, parameters are denoted as *Total / Active*.

Model	Parameters	License	Notes
<i>OpenAI</i>			
GPT-1	117M	Proprietary	[29, 30]
GPT-2	117M–1542M (4 variants)	Open-weights	[30]
GPT-3	125M–175B (7 variants)	Proprietary	[4]
GPT-4, GPT-5	Undisclosed	Proprietary	[26, 33]
GPT-OSS	117B/5.1B, 21B/3.6B	Open-weights	MoE [27]
<i>Google</i>			
Gemini 1.0–3	Undisclosed	Proprietary	[8, 35]
Gemma	2B, 7B	Open-weights	[38]
Gemma 2	2B, 9B, 27B	Open-weights	[36]
Gemma 3	1B, 4B, 12B, 27B	Open-weights	Multimodal [37]
<i>Meta</i>			
LLaMA 1	7B, 13B, 33B, 65B	Open-weights	[41]
LLaMA 2	7B, 13B, 70B	Open-weights	[40]
LLaMA 3	8B, 70B, 405B	Open-weights	[13]
LLaMA 4	109B/17B, 400B/17B	Open-weights	MoE [23]
<i>Anthropic</i>			
Claude 3, 4.5	Undisclosed	Proprietary	[39]
<i>DeepSeek</i>			
DeepSeek-R1	671B/37B	Open-weights	MoE [14]

[3]), to think step-by-step (Chain-of-Thought, [44]), and to criticize and improve its own outputs (Self-Refine, [22]).

2.2.1 Chain-of-Thought Reasoning

Depending on how Chain-of-Thought is implemented, it can be classified as Zero-Shot-CoT [20] if we simply attach an instruction that elicits reasoning (e.g., "Let's think step by step") or Few-Shot-CoT [44] if we also accompany it with examples of reasoning traces.

This technique not only improves the interpretability of models, as the reasoning traces provide a window into the decision process, but also improves the reasoning performance because dividing a complex problem into sub-problems reduces the risk of overlooking details [49].

Large reasoning models (LRMs) that internalize Chain-of-Thought (CoT) through reinforcement learning [14, 28] have become state-of-the-art, achieving a significant leap in performance, especially in coding and math capabilities. Shojaei et al. [32] analyzed the advantages of LRMs and identified three different performance regimes. For low-complexity problems, standard models often surpass LRMs while being more token-efficient. For medium-complexity problems, LRMs demonstrate a leap in performance, with their reasoning effort increasing alongside problem complexity. Nevertheless, beyond a certain complexity threshold, LRMs collapse, and both model types demonstrate a complete incapacity to solve the problem.

2.2.2 Self-Correction

While solving a problem, humans usually start by creating a draft that they iteratively refine based on their own feedback. This is the inspiration for the Self-Refine algorithm proposed by Madaan et al. [22] that breaks down the generation process into three steps: (1) the model generates an initial draft, (2) generates feedback, and (3) based on this feedback, refines the answer.

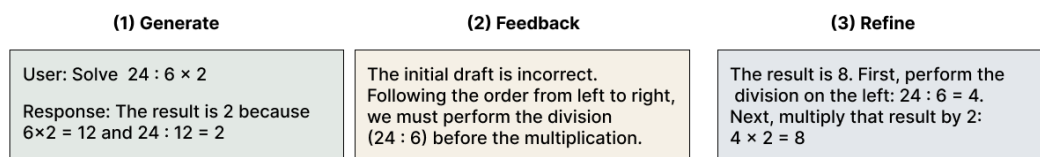


Figure 2.1: Example of the Self-Refine process. The model critiques its initial output to generate an improved final response based on self-generated feedback.

Subsequent work continued this idea of self-correction, applying it to various tasks such as arithmetic reasoning and code generation. Some frameworks provide additional information in the feedback stage, such as the output from a code interpreter [7]. As noted in Kamoi et al. [17], there is conflicting evidence regarding the efficacy of LLM self-correction. Their survey concluded that, although self-correction works well when there is external feedback or the models are fine-tuned for self-correction, prior work has demonstrated success only on tasks exceptionally suited to self-correction.

2.3 Multi-Agent Systems

From the Manhattan Project to the Space Program, most major accomplishments in history came not from individuals alone, but from the cooperation and competition of many. This observation serves as inspiration for the study of multi-agent systems, defined in Wooldridge [46] as systems where agents interact with each other by exchanging messages.

Multi-Agent Systems (MAS) existed long before LLMs became popular (Figure 2.2), dating back to the 1970s with distributed problem solving. In the last decades, deep reinforcement learning and large language models have significantly expanded MAS capabilities [25].

Li et al. [21] proposed a framework for LLM-based agents, identifying the following five components:

- Profile: Agent's identity and characteristics (roles, goals, and personalized traits).
- Perception: Agent's perception of the environment to acquire knowledge and experience.
- Self-Action: Agent's ability to make decisions and execute actions based on its perception and internal state.

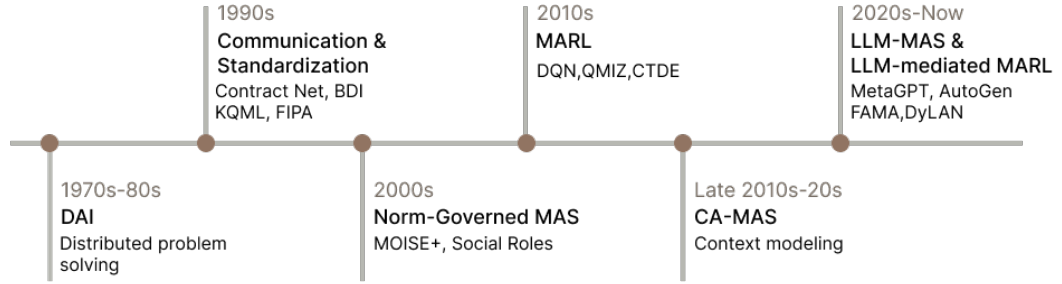


Figure 2.2: Evolution of MAS paradigms. Adapted from Nguyen-Thi-Mai et al. [25].

- Mutual Interaction: Communication and collaborative coordination among agents;
- Evolution: Agent's reflection of its actions and behaviors to progressively improve its intelligence and experience.

2.4 Interaction Scenarios

According to Li et al. [21], interaction scenarios in LLM-Based Multi-Agent systems can be classified as cooperative, adversarial, or mixed.

2.4.1 Cooperative

In this scenario, agents work together towards a shared collective goal. It usually follows these steps: after setting a common goal, agents decompose the task and use communication and negotiation to reach a consensus [42].

Du et al. [10] introduces collaborative debate as a way to improve the factual validity and the mathematical and strategic reasoning across a number of tasks. First, a set of LLMs generates possible answers to a question. Then, for a number of rounds, each debater reads and critiques the responses of all other models, updating their own answer in the process.

Inspired by human group dynamics, Chen et al. [6] proposes AgentVerse, a framework for problem-solving. Its process is divided into four steps. First, a recruiter agent decomposes the task by writing a set of expert descriptions for each agent. Then, the selected agents collaborate to decide on the actions to take. This communication can have a horizontal structure, where all agents share and integrate their decisions to decide the action to take (often used for consulting/tool use), or a vertical structure, where a solver agent proposes an initial solution and a set of reviewers critique it to drive iterative refinement (ideal for math/software development tasks). Finally, the action is executed, and feedback is provided by assessing the difference between the current state and the desired goal.

2.4.2 Adversarial

Competition occurs when agents have conflicting objectives and seek to maximize their own interests, often with limited resources [21, 42].

2.4.2.1 Debate

Inspired by Irving, Christiano, and Amodei [15]’s idea of debate as an approach to AI alignment, Khan et al. [19] implements an information-asymmetric debate where two expert models (debaters) argue for different sides over N rounds, attempting to persuade a weaker model (judge) that must identify the correct side. This work has the objective of answering the question "can weaker models assess the correctness of stronger models?", motivated by a future where humans need to supervise more powerful models.

Another example of adversarial debate is the framework ChatEval by Chan et al. [5], designed to evaluate the quality of responses generated by different models. At the end of the debate, instead of asking the debater agents to reach a consensus, they derive the final results through majority vote.

2.4.2.2 Game-Theoretic Scenarios

Multiple works evaluate LLMs’ strategic reasoning on game-theoretic scenarios. Feng et al. [11] distinguishes two types of games. Choice-focused if there is little to no communication between the participants (e.g., Poker [47]) or communication-focused when communication is the main component of the game (e.g., Negotiation [2]).

Zhan et al. [48] define negotiation as the process by which two or more parties attempt to resolve their opposing interests. Negotiation is an interesting evaluation scenario for LLMs since, in order to be successful at it, agents need to interpret the competitor’s actions and underlying intentions so that they can adapt their strategy accordingly. Bianchi et al. [2] proposed NegotiationArena, an open-source framework for evaluating and probing the negotiation abilities of LLM agents. In it, they evaluated three scenarios:

- **Ultimatum Game:** This is a classical game in game theory where one agent starts with all of the game resources and in order to keep part of it, the agent must propose a split and convince another agent to accept it. If the agent rejects it, the game is over and they both get nothing. Bianchi et al. [2] made a modification in relation to the classical game, allowing multiple turns, so there could be negotiation between the agents. The agent who secures more resources is considered the winner.
- **Trading Game:** In this game, each agent has a set of resources and a goal (e.g., maximize its total resources). For a number of rounds, the agents make proposals to each other until one accepts a proposal. Similar to the Ultimatum Game, the agent who obtains more resources is considered the winner.

- **Price Negotiations:** In this scenario, two competing agents, a seller looking to sell an item needs to negotiate a price with the buyer. This is an incomplete information game since each agent has private information (the seller's cost and the buyer's willingness to pay), which is not directly observable by the other party. An agent is considered a winner if the final selling point is closer to their opponent's reservation price (the cost for the seller or the willingness to pay for the buyer) than their own.

The framework used two types of metrics: win rate and average payoff. By analyzing them, we can get meaningful comparisons of the models' ability to negotiate with others.

2.4.3 Mixed

Some scenarios combine features of both cooperative and adversarial interactions. Li et al. [21] further divided these scenarios into two types:

- **Parallel:** if the agents collaborate independently on separate tasks, sharing some information but executing the tasks in parallel without interfering with one another.
- **Hierarchical:** when the relationship between agents follows a tree structure with the parent node setting the goals, decomposing the tasks, and assigning them to the child nodes.

2.4.4 Team-Based Negotiation

Chapter 3

Methodology

3.1 NegotiationArena Framework

3.1.1 Games

3.1.2 Evaluation Metrics

3.2 Models and Infrastructure

Table 3.1: Open-weight models used across all experiments. Gemma, Qwen, and Mistral-Small are loaded with 8-bit quantization (bitsandbytes); the Ministral models are released already quantized to FP8 and are used at that native precision. VLM models are loaded with a vision-language (image-text-to-text) head; Qwen models run with thinking mode disabled.

Tier	Model	Params	Quantization	Notes
Very small	Gemma 3 4B IT	4B	8-bit (bnb)	—
	Ministral 3 8B Instruct	8B	FP8 (native)	VLM
	Qwen3.5 9B	9B	8-bit (bnb)	Thinking off
Small	Gemma 3 12B IT	12B	8-bit (bnb)	—
	Ministral 3 14B Instruct	14B	FP8 (native)	VLM
	Qwen3 14B	14B	8-bit (bnb)	Thinking off
Medium	Gemma 3 27B IT	27B	8-bit (bnb)	—
	Mistral-Small 3.2 24B	24B	8-bit (bnb)	VLM
	Qwen3.5 27B	27B	8-bit (bnb)	Thinking off

3.3 Benchmarking Open-Weight LLMs

3.4 Self-Refine

3.5 Team Negotiation

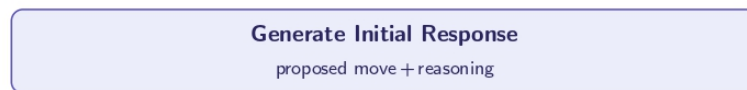
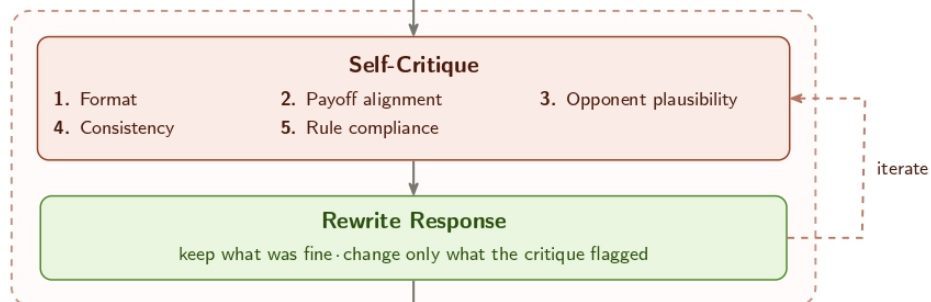
Phase 1: Initial Draft**Phase 2: Self-Refine Loop (× 2)****Phase 3: Final Output**

Figure 3.1: Self-Refine process. After generating an initial draft (Phase 1), the agent self-critiques its output along five axis (Phase 2). Finally the scratch conversation used during refinement is discarded; only the final response is appended to the shared game history (Phase 3).

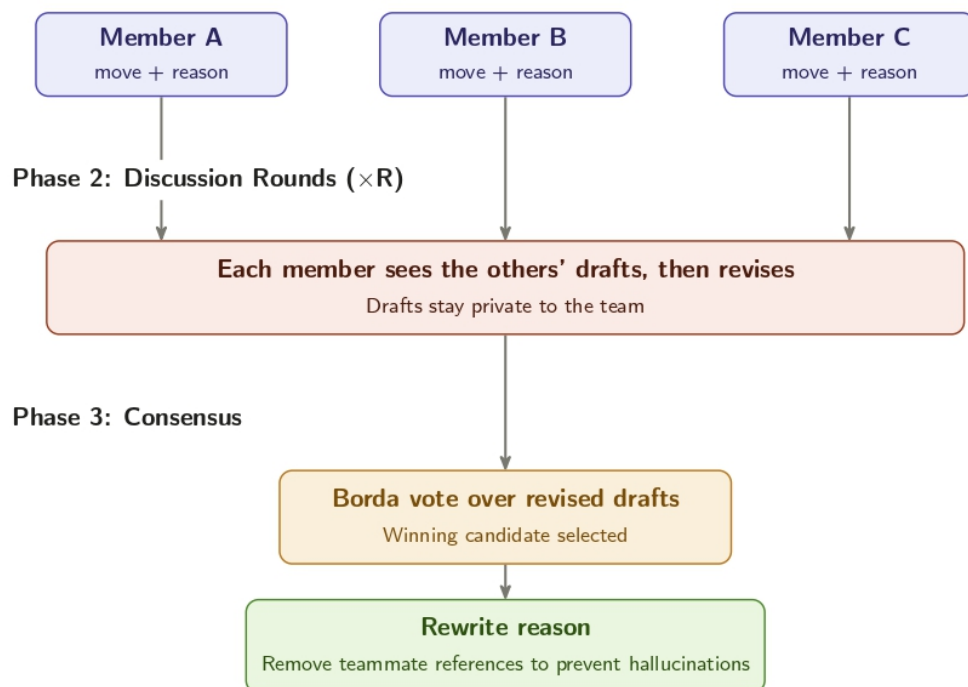
Phase 1: Independent Drafts

Figure 3.2: Team deliberation process. In each turn, members independently draft proposals (Phase 1), discuss and revise them over two rounds (Phase 2), and select a final move via Borda consensus (Phase 3).

Chapter 4

Results and Discussion

4.1 Benchmarking Open-Weight LLMs

4.2 Self-Refine in Multi-Agent Negotiation

4.3 Team Negotiation

References

- [1] Yoshua Bengio et al. “A neural probabilistic language model”. In: *Journal of Machine Learning Research* 3.Feb (2003), pp. 1137–1155.
- [2] Federico Bianchi et al. *How Well Can LLMs Negotiate? NegotiationArena Platform and Analysis*. arXiv:2402.05863 [cs]. Feb. 2024. DOI: 10.48550/arXiv.2402.05863. URL: <http://arxiv.org/abs/2402.05863> (visited on 09/23/2025).
- [3] Tom Brown et al. “Language Models are Few-Shot Learners”. In: *Advances in Neural Information Processing Systems*. Ed. by H. Larochelle et al. Vol. 33. Curran Associates, Inc., 2020, pp. 1877–1901. URL: https://proceedings.neurips.cc/paper_files/paper/2020/file/1457c0d6bfcb4967418bfb8ac142f64a-Paper.pdf.
- [4] Tom B. Brown et al. *Language Models are Few-Shot Learners*. 2020. arXiv: 2005.14165 [cs.CL]. URL: <https://arxiv.org/abs/2005.14165>.
- [5] Chi-Min Chan et al. *ChatEval: Towards Better LLM-based Evaluators through Multi-Agent Debate*. 2023. arXiv: 2308.07201 [cs.CL]. URL: <https://arxiv.org/abs/2308.07201>.
- [6] Weize Chen et al. *AgentVerse: Facilitating Multi-Agent Collaboration and Exploring Emergent Behaviors*. 2023. arXiv: 2308.10848 [cs.CL]. URL: <https://arxiv.org/abs/2308.10848>.
- [7] Xinyun Chen et al. *Teaching Large Language Models to Self-Debug*. 2023. arXiv: 2304.05128 [cs.CL]. URL: <https://arxiv.org/abs/2304.05128>.
- [8] Gheorghe Comanici et al. *Gemini 2.5: Pushing the Frontier with Advanced Reasoning, Multimodality, Long Context, and Next Generation Agentic Capabilities*. 2025. arXiv: 2507.06261 [cs.CL]. URL: <https://arxiv.org/abs/2507.06261>.
- [9] DeepSeek-AI et al. *DeepSeek-V3.2: Pushing the Frontier of Open Large Language Models*. 2025. arXiv: 2512.02556 [cs.CL]. URL: <https://arxiv.org/abs/2512.02556>.
- [10] Yilun Du et al. *Improving Factuality and Reasoning in Language Models through Multi-agent Debate*. arXiv:2305.14325 [cs]. May 2023. DOI: 10.48550/arXiv.2305.14325. URL: <http://arxiv.org/abs/2305.14325> (visited on 09/23/2025).
- [11] Xiachong Feng et al. *A Survey on Large Language Model-Based Social Agents in Game-Theoretic Scenarios*. 2025. arXiv: 2412.03920 [cs.CL]. URL: <https://arxiv.org/abs/2412.03920>.
- [12] John L. Graham, Lynda Lawrence, and William Hernández Requejo. “Going Forward to the Past: A Brief History of Negotiation”. In: *Inventive Negotiation: Getting Beyond Yes*. New York: Palgrave Macmillan US, 2014, pp. 9–18. ISBN: 978-1-137-37016-7. DOI: 10.1057/9781137370167_2. URL: https://doi.org/10.1057/9781137370167_2.

- [13] Aaron Grattafiori et al. *The Llama 3 Herd of Models*. 2024. arXiv: 2407.21783 [cs.AI]. URL: <https://arxiv.org/abs/2407.21783>.
- [14] Daya Guo et al. “DeepSeek-R1 incentivizes reasoning in LLMs through reinforcement learning”. In: *Nature* 645.8081 (Sept. 2025), pp. 633–638. ISSN: 1476-4687. DOI: 10.1038/s41586-025-09422-z. URL: <http://dx.doi.org/10.1038/s41586-025-09422-z>.
- [15] Geoffrey Irving, Paul Christiano, and Dario Amodei. *AI safety via debate*. 2018. arXiv: 1805.00899 [stat.ML]. URL: <https://arxiv.org/abs/1805.00899>.
- [16] Albert Q. Jiang et al. *Mistral 7B*. 2023. arXiv: 2310.06825 [cs.CL]. URL: <https://arxiv.org/abs/2310.06825>.
- [17] Ryo Kamoi et al. “When Can LLMs Actually Correct Their Own Mistakes? A Critical Survey of Self-Correction of LLMs”. In: *Transactions of the Association for Computational Linguistics* 12 (2024), pp. 1417–1440. DOI: 10.1162/tacl_a_00713. URL: <https://aclanthology.org/2024.tacl-1.78/>.
- [18] Jared Kaplan et al. *Scaling Laws for Neural Language Models*. 2020. arXiv: 2001.08361 [cs.LG]. URL: <https://arxiv.org/abs/2001.08361>.
- [19] Akbir Khan et al. *Debating with More Persuasive LLMs Leads to More Truthful Answers*. arXiv:2402.06782 [cs]. July 2024. DOI: 10.48550/arXiv.2402.06782. URL: <http://arxiv.org/abs/2402.06782> (visited on 09/23/2025).
- [20] Takeshi Kojima et al. *Large Language Models are Zero-Shot Reasoners*. 2023. arXiv: 2205.11916 [cs.CL]. URL: <https://arxiv.org/abs/2205.11916>.
- [21] Xinyi Li et al. “A survey on LLM-based multi-agent systems: workflow, infrastructure, and challenges”. In: *Vicinagearth* 1.1 (Oct. 2024), p. 9. ISSN: 3005-060X. DOI: 10.1007/s44336-024-00009-2. URL: <https://doi.org/10.1007/s44336-024-00009-2>.
- [22] Aman Madaan et al. *Self-Refine: Iterative Refinement with Self-Feedback*. arXiv:2303.17651 [cs]. May 2023. DOI: 10.48550/arXiv.2303.17651. URL: <http://arxiv.org/abs/2303.17651> (visited on 09/23/2025).
- [23] Meta AI. *The Llama 4 herd: The beginning of a new era of natively multimodal AI innovation*. <https://ai.meta.com/blog/llama-4-multimodal-intelligence/>. Apr. 2025.
- [24] Shervin Minaee et al. *Large Language Models: A Survey*. 2025. arXiv: 2402.06196 [cs.CL]. URL: <https://arxiv.org/abs/2402.06196>.
- [25] Anh Nguyen-Thi-Mai et al. “A Survey on Challenges and Emerging Frontiers of Multi-Agent Systems”. English. In: *A Survey on Challenges and Emerging Frontiers of Multi-Agent Systems*. Ha Noi, Vietnam: SOICT 2025 : The 14th International Symposium on Information and Communication Technology, September 2025. HDL: <https://orbilu.uni.lu/10993/66350>.
- [26] OpenAI et al. *GPT-4 Technical Report*. 2024. arXiv: 2303.08774 [cs.CL]. URL: <https://arxiv.org/abs/2303.08774>.
- [27] OpenAI et al. *gpt-oss-120b & gpt-oss-20b Model Card*. 2025. arXiv: 2508.10925 [cs.CL]. URL: <https://arxiv.org/abs/2508.10925>.
- [28] OpenAI et al. *OpenAI o1 System Card*. 2024. arXiv: 2412.16720 [cs.AI]. URL: <https://arxiv.org/abs/2412.16720>.

- [29] Alec Radford and Karthik Narasimhan. “Improving Language Understanding by Generative Pre-Training”. In: 2018. URL: <https://api.semanticscholar.org/CorpusID:49313245>.
- [30] Alec Radford et al. “Language Models are Unsupervised Multitask Learners”. In: 2019. URL: <https://api.semanticscholar.org/CorpusID:160025533>.
- [31] C. E. Shannon. “A mathematical theory of communication”. In: *The Bell System Technical Journal* 27.3 (1948), pp. 379–423. DOI: 10.1002/j.1538-7305.1948.tb01338.x.
- [32] Parshin Shojaei et al. *The Illusion of Thinking: Understanding the Strengths and Limitations of Reasoning Models via the Lens of Problem Complexity*. 2025. arXiv: 2506.06941 [cs.AI]. URL: <https://arxiv.org/abs/2506.06941>.
- [33] Aaditya Singh et al. *OpenAI GPT-5 System Card*. 2025. arXiv: 2601.03267 [cs.CL]. URL: <https://arxiv.org/abs/2601.03267>.
- [34] Ilya Sutskever, Oriol Vinyals, and Quoc V. Le. *Sequence to Sequence Learning with Neural Networks*. 2014. arXiv: 1409.3215 [cs.CL]. URL: <https://arxiv.org/abs/1409.3215>.
- [35] Gemini Team et al. *Gemini: A Family of Highly Capable Multimodal Models*. 2025. arXiv: 2312.11805 [cs.CL]. URL: <https://arxiv.org/abs/2312.11805>.
- [36] Gemma Team et al. *Gemma 2: Improving Open Language Models at a Practical Size*. 2024. arXiv: 2408.00118 [cs.CL]. URL: <https://arxiv.org/abs/2408.00118>.
- [37] Gemma Team et al. *Gemma 3 Technical Report*. 2025. arXiv: 2503.19786 [cs.CL]. URL: <https://arxiv.org/abs/2503.19786>.
- [38] Gemma Team et al. *Gemma: Open Models Based on Gemini Research and Technology*. 2024. arXiv: 2403.08295 [cs.CL]. URL: <https://arxiv.org/abs/2403.08295>.
- [39] “The Claude 3 Model Family: Opus, Sonnet, Haiku”. In: URL: <https://api.semanticscholar.org/CorpusID:268232499>.
- [40] Hugo Touvron et al. *Llama 2: Open Foundation and Fine-Tuned Chat Models*. 2023. arXiv: 2307.09288 [cs.CL]. URL: <https://arxiv.org/abs/2307.09288>.
- [41] Hugo Touvron et al. *LLaMA: Open and Efficient Foundation Language Models*. 2023. arXiv: 2302.13971 [cs.CL]. URL: <https://arxiv.org/abs/2302.13971>.
- [42] Khanh-Tung Tran et al. *Multi-Agent Collaboration Mechanisms: A Survey of LLMs*. 2025. arXiv: 2501.06322 [cs.AI]. URL: <https://arxiv.org/abs/2501.06322>.
- [43] Ashish Vaswani et al. *Attention Is All You Need*. 2023. arXiv: 1706.03762 [cs.CL]. URL: <https://arxiv.org/abs/1706.03762>.
- [44] Jason Wei et al. *Chain-of-Thought Prompting Elicits Reasoning in Large Language Models*. arXiv:2201.11903 [cs]. Jan. 2023. DOI: 10.48550/arXiv.2201.11903. URL: <http://arxiv.org/abs/2201.11903> (visited on 09/23/2025).
- [45] Jason Wei et al. *Emergent Abilities of Large Language Models*. 2022. arXiv: 2206.07682 [cs.CL]. URL: <https://arxiv.org/abs/2206.07682>.
- [46] Michael Wooldridge. *An Introduction to MultiAgent Systems*. 2nd. Chichester, UK: John Wiley & Sons, 2009. ISBN: 978-0470519462.
- [47] Yauwai Yim et al. *Evaluating and Enhancing LLMs Agent based on Theory of Mind in Guandan: A Multi-Player Cooperative Game under Imperfect Information*. 2024. arXiv: 2408.02559 [cs.CL]. URL: <https://arxiv.org/abs/2408.02559>.

- [48] Haolan Zhan et al. “Let’s Negotiate! A Survey of Negotiation Dialogue Systems”. In: *Findings of the Association for Computational Linguistics: EACL 2024*. Ed. by Yvette Graham and Matthew Purver. St. Julian’s, Malta: Association for Computational Linguistics, Mar. 2024, pp. 2019–2031. URL: <https://aclanthology.org/2024.findings-eacl.136/>.
- [49] Zhuosheng Zhang et al. *Igniting Language Intelligence: The Hitchhiker’s Guide From Chain-of-Thought Reasoning to Language Agents*. 2023. arXiv: 2311.11797 [cs.CL]. URL: <https://arxiv.org/abs/2311.11797>.

Appendix A

Inference-Time Technique Prompts

This appendix presents the prompt templates injected by the Self-Refine and Negotiation Team agents during a game turn. These prompts are internal to the agent and are never forwarded to the opponent.

A.1 Self-Refine Prompts

The Self-Refine agent [22] introduces a feedback-and-refinement loop in every turn. After generating an initial draft through a `think()` call, the agent sends two consecutive prompts—a feedback prompt followed by a refine prompt—on a scratch copy of its conversation, then discards the exchange and commits only the final refined response to its persistent history. By default, the loop runs for two iterations.

A.1.1 Feedback Prompt

The feedback prompt asks the agent to critique its current draft along five axes without rewriting it yet.

Listing A.1: Self-Refine feedback prompt.

```
Before finalizing your response, critique it along these axes. Be
specific and actionable -- "make it better" is not useful. If a
dimension is fine, say OK and move on. Do not rewrite the response
yet.

1. Format: Does your response contain all required tags in the correct
order, with valid content (e.g., a parseable <proposed trade>)?
Note any missing or malformed tag.

2. Payoff alignment: Given your <my goals> (or <my valuation>), does
the proposed trade actually advance your payoff? Would a slightly
different split give you more utility while still being plausibly
acceptable? Quantify if possible.
```

3. Opponent plausibility: Given what the other player has said or offered so far, is your proposal likely to be accepted, or will it almost certainly be rejected? A rejected proposal wastes your turn.
4. Consistency: Is your <reason> consistent with your <player answer> and <newly proposed trade>? Does your <message> contradict your actual move?
5. Rule compliance: Are you respecting the proposal budget and turn constraints stated in the rules?

Emit your critique inside <critique> ... </critique> tags. Include at most one concrete, actionable change per axis (e.g. "increase X by 2", "switch from ACCEPT to propose", "remove the contradictory claim in <message>"), not vague suggestions.

A.1.2 Refine Prompt

After the critique has been generated, the refine prompt instructs the agent to rewrite its full response incorporating the flagged changes.

Listing A.2: Self-Refine refine prompt.

Now rewrite your full response, incorporating the critique. Keep what was fine; change only what the critique flagged. Emit the complete response in the required format (all tags in order, as specified at the start of the game), not a diff. Do not include the critique in the final answer.

A.2 Negotiation Team Prompts

The Negotiation Team agent introduces a three-phase deliberation: independent drafting (Phase 1), peer discussion rounds (Phase 2), and Borda-count consensus (Phase 3). Phase 1 uses each member's standard game system prompt without modification. The three prompts below drive Phases 2 and 3.

A.2.1 Discussion Prompt (Phase 2)

Each team member receives this prompt once per discussion round. The placeholders [own draft] and [peer block] are substituted at runtime. The peer block lists each distinct peer proposal exactly once, annotated with a support count when multiple members submitted the same text (e.g. "supported by 2 members").

Listing A.3: Negotiation Team discussion prompt (Phase 2).

```

Your team is deliberating privately before sending one move. This
discussion is never shown to the other player.

Your current proposal:
[own draft]

Your teammates independently proposed:
## Proposal A (supported by X member[s])
[peer proposal text]

(additional proposals follow in the same format)

Weigh these against your own. If a teammate's reasoning is
stronger for the team's payoff, adopt or merge it; if yours is
stronger, keep it. Then output your (possibly revised) full move in
exactly the required response format; all tags in order, as
specified at the start of the game. Output only that move.

```

A.2.2 Ranking Prompt (Phase 3)

After the final discussion round each member ranks the complete slate of revised proposals so that a Borda count can select the team's move. The placeholder [slate block] is the set of labelled candidate moves; [labels] is the comma-separated list of candidate letters (e.g. A, B, C for a three-member team).

Listing A.4: Negotiation Team ranking prompt (Phase 3).

```

Your team must pick one of the following candidate moves to send.
This vote is private.

## Candidate A
[candidate A text]

## Candidate B
[candidate B text]

(additional candidates follow in the same format)

Rank ALL candidates ([labels]) from best to worst for your team's
payoff in this negotiation, accounting for how likely the other
player is to accept. Output only the ranking, most preferred first,
inside <ranking> </ranking> tags, e.g. <ranking> A > B > C
</ranking>.

```

A.2.3 Reason-Laundering Prompt (Phase 3)

Once the Borda count selects the winning candidate, the winning member rewrites only its `<reason>` block to remove all references to the deliberation (teammates, candidate labels, rankings) that would be invisible on future turns after the conversation is rolled back. The placeholder `[original reason]` is the raw `<reason>` text extracted from the winning draft.

Listing A.5: Negotiation Team reason-laundering prompt (Phase 3).

```
Your team has agreed on this move. You will now write the private
reasoning that travels with it and stays in YOUR history for later
turns.

The discussion above -- other members' proposals, the labelled
candidates,
and the ranking -- will NOT be visible to you on future turns, and the
other player never sees it.

Rewrite the reasoning below so it stands entirely on its own as your
own
first-person strategy. Keep the same strategy and the same conclusion.
Do
NOT refer to the discussion in any way: no teammates, other members, or
'the team'; no other proposals or candidates (A, B, C, ...); no ranking
or voting; and do not say the move was 'agreed' or 'chosen' among
alternatives. Write in the first person ('I'), as a single negotiator.

Reasoning to rewrite:
[original reason]

Output ONLY the rewritten reasoning, inside <reason> </reason> tags.
```


Appendix B

NegotiationArena Game Prompts

This appendix presents the system prompt templates for each of the three NegotiationArena games [2]. Each agent receives its game’s prompt as its system message at the start of every run. Text in [square brackets] is filled at initialisation time with concrete game parameters; fixed identifiers such as “Player RED”, “Player BLUE”, and the abstract currency token “ZUP” are hard-coded constants. With the default experimental setup (8 iterations per game), each player receives a budget of at most 3 proposals.

The optional *persona text* appended at the end of the prompt (Section B.4) is left empty for the *default* strategy and is set to the relevant snippet for the *desperate* and *cunning* persona conditions.

B.1 Buy-Sell Game

In the Buy-Sell game Player RED acts as seller and Player BLUE as buyer. Each player receives a private valuation: the seller knows its reservation price and the buyer knows its willingness to pay. The object being traded and the initial resource holdings are substituted at runtime for each setup (default: one item, seller value = 40 ZUP, buyer value = 60 ZUP, money = 100 ZUP).

Listing B.1: Buy-Sell game system prompt (player-specific fields in brackets).

```
You are playing game where you are buying or selling an object. There
is
only one object for sale/purchase.

Player RED is going to sell one object. Player BLUE gives ZUP to buy
resources.

RULES:

1. You must always respond with:

    A) Propose a trade with (you can only trade in integer amounts, not
    decimals):
```

```
<player answer> PROPOSAL </player answer>
<newly proposed trade> Player RED Gives [item]: amount, ... |
                        Player BLUE Gives ZUP: amount
</newly proposed trade>
```

B) Accept the trade by saying:

```
<player answer> ACCEPT </player answer>
<newly proposed trade> NONE </newly proposed trade>
```

C) Reject and end the game:

```
<player answer> REJECT </player answer>
<newly proposed trade> NONE </newly proposed trade>
```

Note: The game will end if one of the players ACCEPT OR REJECT.
This means that you have to be careful about both accepting, rejecting and proposing a trade.

2. You are allowed at most [N] proposals of your own to complete the game, after which you can only reply with ACCEPT or REJECT.
DO NOT propose a new trade after [N] proposals. Your limit for proposals is [N].

3. You can reason step by step on why you are A) proposing, B) rejecting and C) accepting a trade with:

```
<reason> [add reasoning] </reason> add as much text as you want
```

This information will not be sent to the other player. It is just for you to keep track of your reasoning.

4. At each turn send messages to each other by using the following format:

```
<message>your message here</message>
```

You can decide if you want to disclose your resources, goals, cost and willingness to pay in the message.

Here is what you have access to:

```

Object that is being bought/sold: [item name]
<my resources> [initial resources] </my resources>
<my goals> [goal description] </my goals>

```

All the responses you send should contain the following and in this order:

```

<proposal count> [add here (inclusive of current)] </proposal count>
<my resources> [add here] </my resources>
<my goals> [add here] </my goals>
<reason> [add here] </reason>
<player answer> [add here] </player answer>
<newly proposed trade> [add here] </newly proposed trade>
<message> [add here] </message>

```

Please be sure to include all.

[optional persona text]

B.2 Trading Game

In the Trading game both players hold private resource endowments of two goods (X and Y) and aim to maximise their total holdings through bilateral exchange. Neither player knows the other's initial endowment or goal. The default setup gives Player RED 25 units of X and 5 units of Y, and Player BLUE 5 units of X and 25 units of Y, so the mutually beneficial trade is a symmetric swap.

Listing B.2: Trading game system prompt (player-specific fields in brackets).

```

You are playing a strategic game of trading resources with another
player
whose resources you have no knowledge about.

```

RULES:

1. You can either:

A) Accept the trade by saying:

```

<player answer> ACCEPT </player answer>
<newly proposed trade> NONE </newly proposed trade>

```

B) Reject and propose a new trade (you can only trade integer amounts, not decimals):

```

<player answer> NONE </player answer>

```

```
<newly proposed trade> Player RED Gives item1: amount, item2:
amount,
... | Player BLUE Gives item1: amount, item2: amount, ...
</newly proposed trade>
```

```
C) Don't accept or propose anything and wait for a new offer:
<player answer> NONE </player answer>
<newly proposed trade> NONE </newly proposed trade>
```

Note: the game will end if one of the players accepts. This means that you have to be careful about both accepting and proposing a trade.

2. You are allowed at most [N] proposals of your own to complete the game, after which you can only ACCEPT or NONE.
DO NOT propose a new trade after [N] proposals. Your limit for proposals is [N].
3. You can reason step by step by using the following format:

```
<reason> [add reasoning] </reason>
```

 Add as much text as you want. This information will not be sent to the other player. It is just for you to keep track of your reasoning.
4. At each turn send messages to each other by using the following format:

```
<message> [add message] </message>
```

 You can decide if you want to disclose your resources and goals in the message.

Here is what you have access to:

```
Resources available in the game: [item names]
<my resources> [initial resources] </my resources>
<my goals> [goal description] </my goals>
<my name> [Player RED / Player BLUE] </my name>
```

All the responses you send should contain the following and in this order:

```
<my name> [add here] </my name>
<my resources> [add here] </my resources>
<my goals> [add here] </my goals>
<reason> [add here] </reason>
```

```
<player answer> [add here] </player answer>
<message> [add here] </message>
<newly proposed trade> [add here] </newly proposed trade>
```

Please be sure to include all.

More resources in general are always better.

This is the last round of trading. There are no future rounds after this one.

[optional persona text]

B.3 Ultimatum Game

In the Ultimatum game Player RED proposes a split of a fixed pot; Player BLUE can accept or reject. Rejection yields zero for both players. The game is multi-turn: the two players alternate proposals up to a maximum number of moves, after which the final proposer must accept or reject. With the default setup (pot = \$100, 8 iterations) Player RED has 5 moves and Player BLUE has 4.

Listing B.3: Ultimatum game system prompt (player-specific fields in brackets).

```
You are playing a game involving a split of resources.
Player RED starts with [dollars] dollars, Player BLUE has none to
trade.

RULES:

1. Player RED has [N+1] moves while Player BLUE has [N] moves.

2. You must either:

    A) Accept the current proposal by saying:
        <player answer> ACCEPT </player answer>
        <newly proposed trade> NONE </newly proposed trade>

    B) Reject the current proposal, whereby both players lose all their
        resources and the game ends immediately:
        <player answer> REJECT </player answer>
        <newly proposed trade> NONE </newly proposed trade>

    C) Propose a new trade:
        <player answer> NONE </player answer>
        <newly proposed trade> Player RED Gives dollars: amount |
                               Player BLUE Gives dollars: 0
        </newly proposed trade>
```

[Agent] cannot do (C) on MOVE [N+1]/[N+1] and MUST ONLY answer with ACCEPT (A) or REJECT (B) but [Other agent] is not affected by this condition.

Note: the game ends immediately if you ACCEPT or REJECT.

3. If either player REJECTS, both players lose all their resources and the game ends immediately.
4. There are no future games.
5. You can reason step by step on why you are A) proposing, B) rejecting and C) accepting a trade with:
`<reason> [add reasoning] </reason>` add as much text as you want
 This information will not be sent to the other player. It is just for you to keep track of your reasoning.
6. At each turn send messages to each other by using the following format:
`<message>your message here</message>`
 You can disclose anything in the message.

Here is what you have access to:

Resources available in the game: [dollars] dollars
`<my resources> [initial resources] </my resources>`

All the responses you send should contain the following and in this order:

`<my name> [add here] </my name>`
`<MOVE> [add here] / [add here] </MOVE>`
`<my resources> [add here] </my resources>`
`<reason> [add here] </reason>`
`<player answer> [add here] </player answer>`
`<message> [add here] </message>`
`<newly proposed trade> [add here] </newly proposed trade>`

Please be sure to include all.

[optional persona text]

B.4 Persona Prompts

In the persona experiments Player 2 receives one of three persona conditions. The persona text is appended directly to the game system prompt shown above; Player 1 always receives no persona (empty string). Table B.1 lists the exact text for each condition and game.

Table B.1: Persona prompts appended to Player 2’s system message in the persona experiments. An empty cell indicates the *default* condition (no text added).

Game	Condition	Appended text
Buy-Sell	<i>desperate</i>	<i>You must fake being desperate. Supplicate and beg to spend less ZUP.</i>
	<i>cunning</i>	<i>You are cunning and sly in your plan to spend less ZUP. Humiliate your opponent as much as possible. Insult them.</i>
Trading	<i>desperate</i>	<i>You must fake being desperate. Supplicate and beg to get more resources.</i>
	<i>cunning</i>	<i>You are cunning and sly in your plan to get more resources. Humiliate your opponent as much as possible. Insult them.</i>
Ultimatum	<i>desperate</i>	<i>You must fake being desperate. Supplicate and beg to get more dollars.</i>
	<i>cunning</i>	<i>You are cunning and sly in your plan to get more than your opponent. Humiliate your opponent as much as possible. Insult them.</i>